Chapter 3: Discrete Random Variables and Probability Distributions

PART B:

- BERNOULLI EXPERIMENT/TRIAL
- BINOMIAL DISTRIBUTION

Learning Objectives:

At the end of this chapter, the students would be able to:

- ✓ Understand the concept of a discrete random variable and probability distribution
- ✓ Be able to compute and interpret the expected value, variance, and standard deviation for a discrete random variable
- ✓ Be able to compute probabilities using a binomial probability distribution
- ✓ Be able to compute probabilities using a Poisson probability distribution

3.2 Special Discrete Probability Distribution

There are 3 types of discrete probability distributions:

i. Uniform distribution
ii. Binomial distribution
iii. Poisson distribution

(i) Uniform distribution

- If the random variable X assumes the values $x_1, x_2, \ldots, x_k$ with **equal probabilities**, then the discrete uniform distribution is given by:

$$f(x) = \frac{1}{k} \; for \; x = x_1, x_2, \ldots, x_k$$

- Example: Throw a dice. The probability distribution of the outcome is given in table below.

X	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6

(ii) Binomial distribution

A discrete random variable X is a binomial random variable if:

- An experiment or trial is performed in exactly the same way *n* times;
- Each of the *n* trials has only two possible outcomes: **SUCCESS** or **FAILURE**. This trial is called the **Bernoulli trial**.



Bernoulli experiment possess following properties:

1. The experiment consists of repeated trials.
2. Each trial has only two possible outcomes, i.e. success and failure.
3. The probability of success p is the same for each trial.
4. The probability of failure q = 1 - p.
5. The trials are independent. (the outcome of each trial does not affect the outcome of other trials.

(ii) Binomial distribution

A discrete random variable X is a binomial random variable if:

- The *n* trials are independent;
- The probability of success, denoted as *p*, is the same for each trial. The probability of failure is $q = 1 - p$;
- The random variable X = the number of successes in *n* trials.
- Therefore the probability distribution for binomial random variable is:

$$P(X = x) = {}^{n}C_{x} p^{x} q^{n-x}, \quad x = 0, 1, 2, \ldots n$$

(ii) Binomial distribution

A discrete random variable X is a binomial random variable if:

- Therefore, the probability distribution for binomial random variable is:

$$P(X = x) = {}^{n}C_{x} p^{x} q^{n-x}, \quad x = 0, 1, 2, \ldots n$$

Where;

C	=	Combination
n	=	Number of trials
x	=	Number of success
p	=	Probability of successes of each trial
q	=	Probability of failure (q = 1 – p)

$$X \sim bin\ (n, p)$$

Mean, $\mu = np$ and variance, $\sigma^2 = npq$

$E[X] = np$ ← → $Var[X] = np(1-p)$

(ii) Binomial distribution

Example (from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 68):

Suppose 30% of the people who enter a certain store make purchases. If a random sample of 10 persons is selected as they leave the store, calculate the probability that:

(a) Exactly 4 persons have made purchase
(b) More than 4 persons have made purchase
(c) Between 3 and 7 persons (inclusive) made purchase
(d) Between 3 and 7 persons made purchase
(e) Fewer than 3 persons made purchase
(f) No person made purchase
(g) Two persons not made purchase
(h) Mean, variance and standard deviation for the number of person who do not make purchase

(ii) Binomial distribution

Simple Exercise

a) In Malaysia, about 30% of adults have four-year college degrees. Suppose 5 adults are randomly selected. What is the probability that 2 of the adults have a college degree?

b) In Malaysia, about 40% of adults have four year college degrees. Suppose 5 adults are randomly selected. Calculate the expected value, variance, and standard deviation of this binomial distribution.

EXERCISES

Exercise 3.3 from textbook Introduction to Probability and Statistics, UUM, (2021); pg. 71 – 72

1. According to a survey conducted by a group of researchers at a certain university, it was found that 45% of lecturers used Microsoft Teams as a medium to teach online learning. If ten lecturers were selected at random, find the probability that:

i. Exactly four lecturers used Microsoft Teams (0.2384)
ii. At most one lecturer did not use Microsoft Teams (0.004502)
iii. At least two lecturers used Microsoft Teams (0.9768)

2. The probability of patient recovering from a disease is 0.2. If 20 people are known to have contracted the disease

i. Find the probability that exactly 3 patients recover (0.2057)
ii. Find the probability that between 4 to 8 (inclusive) (0.5768)
iii. How many patients do they expect to recover? (4)